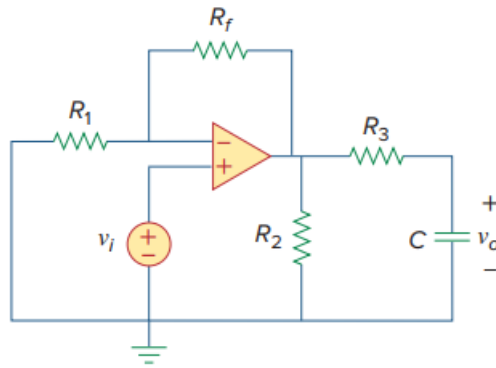


Problem

Obtain the step response $v_o(t)$ for the circuit in Fig. 7.62. Let $v_i = 9u(t)$ V, $R_1 = 20$ k Ω , $R_f = 40$ k Ω , $R_2 = R_3 = 10$ k Ω , $C = 2$ μ F.

Fig. 7.62:



Step-by-step solution

Step 1 of 8

Refer Figure 7.62 from the textbook.

Temporarily remove the capacitor and find the Thevenin equivalent at capacitor terminals.

To obtain V_{Th} across the capacitor terminals, redraw the circuit as shown in Figure 1.

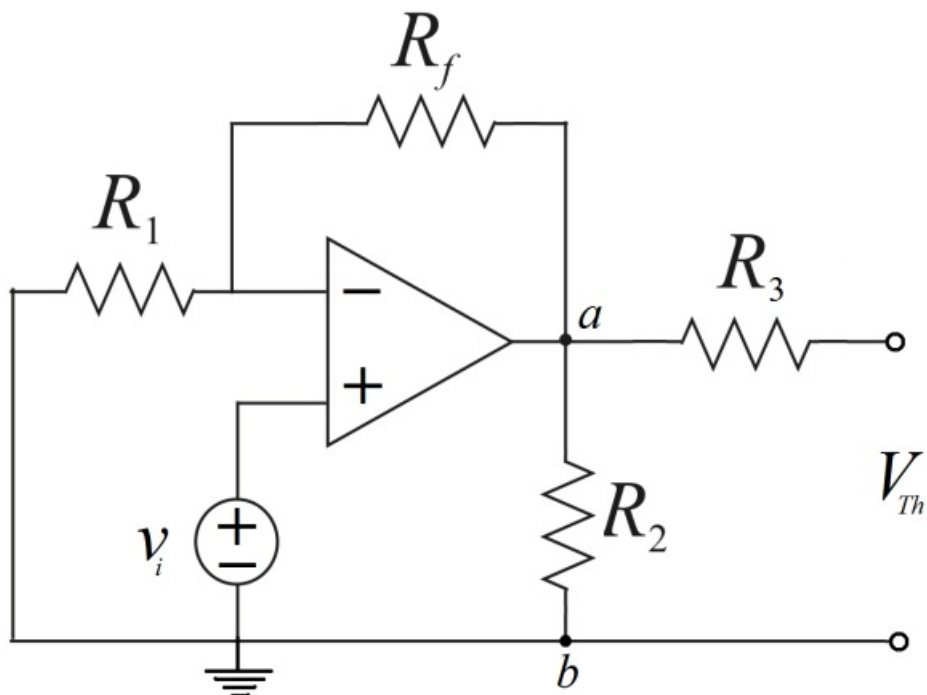


Figure 1

Step 2 of 8

This circuit diagram is operating non-inverting configuration. Hence the voltage V_{ab} is,

$$V_{ab} = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Substitute 4.5 V for v_i , $20 \text{ k}\Omega$ for R_1 , and $40 \text{ k}\Omega$ for R_f .

$$\begin{aligned} V_{ab} &= \left(1 + \frac{40 \times 10^3}{20 \times 10^3}\right) 4.5 \\ &= (1 + 2) 4.5 \\ &= 13.5 \text{ V} \end{aligned}$$

From Figure 1, the voltage V_{Th} is,

$$\begin{aligned} V_{Th} &= V_{ab} \\ &= 13.5 \text{ V} \end{aligned}$$

To obtain R_{Th} across capacitor terminals, the circuit can be represented as shown in Figure 2.

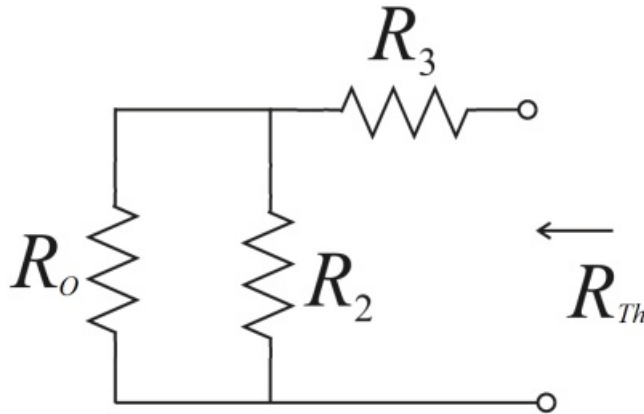


Figure 2

Here, R_o is the output resistance of the op-amp.

Since the op amp is ideal, the output resistance is,

$$R_o = 0$$

Step 3 of 8

Now, the circuit can be redrawn as shown in Figure 3.

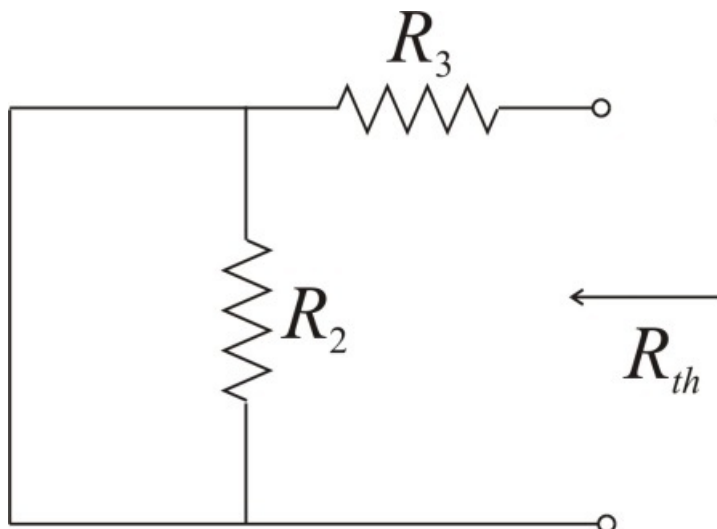


Figure 3

Step 4 of 8

In Figure 3, R_2 is short circuited. Thus, the value of R_{Th} is,

$$\begin{aligned} R_{Th} &= R_3 \\ &= 10 \text{ k}\Omega \end{aligned}$$

The Thevenin equivalent of the circuit is shown in Figure 4.

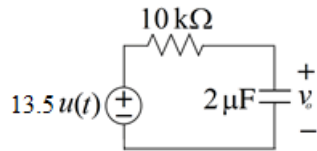


Figure 4

Step 5 of 8

The step function can be defined as follows:

$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t > 0 \end{cases}$$

Step 6 of 8

For $t < 0$ capacitor becomes open circuited and the resultant circuit is shown in Figure 5.

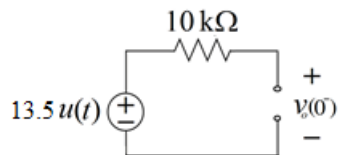


Figure 5

Step 7 of 8

From Figure 5, calculate the value of $v_o(0^-)$.

$$v_o(0^-) = 0$$

Capacitor voltage cannot change instantaneously.

$$\begin{aligned} v_o(0^-) &= v_o(0^+) \\ &= v_o(0) \\ &= 0 \text{ V} \end{aligned}$$

For $t > 0$, voltage source comes into action and circuit reached to steady state. Then, the capacitor again becomes open circuited. The resultant circuit is shown in Figure 6.

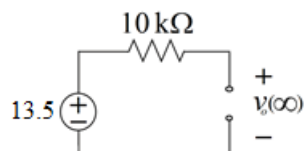


Figure 6

Step 8 of 8

From Figure 6, calculate the value of $v_o(\infty)$.

$$v_o(\infty) = 13.5 \text{ V}$$

Calculate time constant.

$$\begin{aligned}\tau &= R_{Th}C \\ &= (10 \times 10^3)(2 \times 10^{-6}) \\ &= 20 \times 10^{-3} \\ &= 0.02 \text{ sec}\end{aligned}$$

The capacitor voltage equation is,

$$\begin{aligned}v_o(t) &= v_o(\infty) + [v_o(0) - v_o(\infty)]e^{-\frac{t}{\tau}} \\ &= 13.5 + (0 - 13.5)e^{-\frac{t}{0.02}} \\ &= 13.5(1 - e^{-50t}) \quad \text{for } t > 0 \\ &= 13.5(1 - e^{-50t})u(t) \text{ V}\end{aligned}$$

Thus, the step response $v_o(t)$ is,

$$\boxed{v_o(t) = 13.5(1 - e^{-50t})u(t) \text{ V}}.$$